

NEKHA BOSE

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SUMMARY

Result-driven Sr. Software Engineer with 5+ years of experience in developing scalable software solutions and leveraging data for business insights. Currently pursuing a Master's degree in Information Technology and Management at Illinois Institute of Technology. Experienced in building real-time processing systems, and visualizing data for key stakeholders. Certified Azure AI Engineer Associate, showcasing expertise in Azure AI technology and data-driven solutions.

EDUCATION

Illinois Institute of Technology, Chicago, IL

January 2024 - December 2025

Information Technology and Management - GPA 3.88/4.0

APJ Abdul Kalam Technological University, Kerala

August 2016 - September 2020

Information Technology

WORK EXPERIENCE

Lead Software Developer - BreatheIT Solutions Inc

December 2024 - Present

Lifestyle Technology platform Development

- Engineered an AI powered **Lifestyle platform** with a **Node.js** backend implementing RESTful APIs with optimized query handling and responsive frontend components built using **React.js** for the business portal and **React Native** for the mobile application for both **Android and IOS**, collaborating closely with designers.
- Implemented **JWT** authentication using an access and refresh token mechanism, ensuring secure user sessions, automated token renewal, and robust access control and also Implemented **WebSockets** for real-time notifications. Designed and managed **50+ PostgreSQL** tables across multiple databases ensuring efficient data retrieval and following best practices.
- Deployed **100+** endpoints in GCP VMs using docker and configured the endpoints using Nginx for load balancing.
- Mentored **15+ developers** ensuring best coding practices and timely delivery.
- Lead **Scrum** calls, followed Agile methodologies for continuous integration and deployment, ensuring 30% faster feature rollouts.
- Performed Unit Testing and Integration testing utilizing Postman for rigorous API testing, reducing debugging cycles and ensuring seamless request-response handling.

Sr. Software Engineer - Infosys Limited

March 2021 - December 2023

E-commerce platform Development

- Led** the creation of a scalable E-Commerce platform using **Java (Spring Boot)** for backend and React.js for frontend, ensuring robust performance. Successfully supported over **10 Million** concurrent users with **99.99%** uptime using Docker and Kubernetes, directly driving a 20% increase in user satisfaction and retention.
- Integrated MongoDB and Postgres to enhance backend data retrieval and storage, optimizing query performance and reducing response times by 40%, which significantly improved overall system efficiency and user experience.
- Conducted risk assessments and provided timely technical reports to stakeholders, improving process transparency. Participated in **Scrum and Agile**, facilitating continuous improvement and timely delivery.

Data Automation and ETL Optimization for Nike

- Spearheaded the implementation of ETL processes using **Autosys, Apache Airflow, and AWS**, reducing data processing time by 40% and improving data retrieval speed by 30%.
- Collaborated with business analysts to design and build data visualizations and reports using Tableau, providing stakeholders with clear, actionable insights, improving decision-making efficiency by **30%** and increasing team productivity by **25%**.
- Integrated **Spark** jobs with S3 for raw data ingestion and Amazon **Redshift** for downstream analytics, implementing automated workflows in Airflow
- Worked closely with product managers to ensure alignment between technical solutions and business objectives.

Software Engineer - Cogniz Info Tech

June 2017 - December 2019

Interactive Dashboard for Sales Analytics

- Developed an interactive dashboard using React.js for sales analytics, improving user engagement by 20% through optimized typography, enhancing readability and user interaction.
- Integrated the frontend with a Node.js/Express backend to fetch and process data from a **MongoDB** database. Utilized RESTful APIs to ensure seamless data flow, providing users with up-to-date sales insights and performance metrics, resulting in a 30% improvement in sales decision-making efficiency.

PROJECTS

Travel Tracker: MVVM Android App with Firebase, Gemini AI & Interactive Map

January 2025 - May 2025

- Employed **MVVM** with Hilt for dependency injection, LiveData & ViewModel, and **Kotlin** Coroutines to integrate Firebase Authentication (Email/Password) and Realtime Database, using JSON serialization for efficient, private trip data handling.
- Programmed RESTful APIs with Google's Gemini AI SDK to generate personalized travel itineraries based on user preferences and real-time location data; improved user engagement by 6x., and used Jetpack Lifecycle Components alongside Geospatial Coordinates & Geocoding APIs to display trips on Google Maps with accurate markers.

- Architected complex entity relationships with ORM (JPA) for a multi-user e-commerce platform; ensured customer data security and reliability by implementing a robust authentication and authorization process.
- Fashioned reusable composite UI components, cutting front-end development time by 30% and ensuring the UI was consistent with brand guidelines during the project.
- Implemented modular code using MVC, DAO layers, and configured custom security realms on Payara Server; improved system maintainability

SKILLS

- **Front-end:** React.js, Redux, React Native, Angular, HTML5, CSS3, SCSS, JavaScript, TypeScript, jQuery, D3.js, FlutterFlow, Bootstrap, web development
- **Back-end & APIs:** Node.js/Express, Python, Java (Spring Boot), REST APIs, GraphQL, Maven, Microservices, Serverless (AWS Lambda), JavaFX
- **DevOps:** Docker, containerization, Kubernetes, CI/CD (Jenkins, GitHub Actions), Nginx, PM2
- **Data & Cloud Technologies:** ETL processes with Autosys, Apache Airflow, AWS Glue, Data transformations in Teradata, Tableau, Real-time Data Processing, Model Deployment, AWS (EC2, Lambda), GCP (Compute Engine, App Engine, Cloud Run), Kafka.
- **Databases:** SQL(Postgres, SQLite), NoSQL(MongoDB), Amazon S3, Google Cloud Storage
- **Testing & Tools:** Jasmine, JUnit, Postman, Swagger, Webpack, Babel
- **Methodologies:** Scrum, Agile, Kanban, WaterFall, SaaS, B2B, B2C, SDLC, Feature Grooming, Story Point Estimation, Code Reviews.
- **Design & Collaboration:** Figma, Sketch, Jira, Confluence, InVision, Wireframe Translation, UI/UX Implementation

ACHIEVEMENTS

- **Microsoft Certified: Azure AI Engineer Associate** from Microsoft, demonstrating expertise in Azure AI technology.
- Awarded the Best Student Project (Braille Converter) in the University for the year 2019-20.
- Presented a paper in the SERB sponsored National Conference on recent Developments in Computational Method for renewable energy.